Adams Jabali Momanyi

Software Engineer | Data Scientist | Automation Engineer

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Summary

Software Engineer with hands-on experience developing **backend systems, RESTful APIs, automation tools, and full-stack web applications**. Experienced in **Python, JavaScript, Node.js, Flask, FastAPI, PostgreSQL, Docker, and Git** through internships and personal projects. Passionate about building **scalable, secure software solutions** and eager to contribute to **innovative fintech products** while continuously learning from experienced engineering teams. Recently completed a BSc in Data Science from Meru University of Science and Technology.

Technical Skills

- **Languages:** Python (Expert), JavaScript/TypeScript (Proficient), SQL (Advanced), R
- **Backend & Software Engineering:** Node.js, Flask, FastAPI, RESTful APIs, PostgreSQL, Docker, Git, Scalable Backend Systems, Secure API Design
- **Data Science & Machine Learning:** TensorFlow, Scikit-learn, PyTorch, NLP, Predictive Modeling, Collaborative Filtering, Neural Networks
- **Frontend & Visualization:** React.js, Tailwind CSS, Streamlit, Power BI, Matplotlib, Seaborn
- **Cloud & Tools:** AWS, GCP, Linux, Apache Airflow, Jupyter Notebook, VS Code, ETL Pipelines

Experience

Software Engineer Intern / Automation Engineer | Statsspeak Group Ltd | Nairobi, Kenya

May 2026 – Present (3-month contract)

- Developed and implemented **Python-based automation scripts** and workflows to **streamline data pipelines**, improving efficiency in data ingestion and processing.

- Assisted in the design and deployment of **machine learning models**, including testing and evaluation using appropriate metrics.
- Conducted **exploratory data analysis (EDA)** and statistical summarization on large datasets to identify trends and actionable insights.
- Created **data visualizations** and prepared reports for both technical and non-technical stakeholders, translating complex data into clear business narratives.
- Gained exposure to **Apache Airflow** and other orchestration frameworks for managing automated data workflows.

ICT Intern | KUCCPS (Kenya Universities and Colleges Central Placement Service) | Nairobi, Kenya

April – August (2025)

- Assisted in **data cleaning, validation, and processing** of student placement records, ensuring data accuracy and integrity.
- Supported **database integrity** and secure data handling practices, contributing to robust data management.
- Provided **ICT troubleshooting and user support**, resolving technical issues for internal users.
- Gained exposure to **government data governance standards** and best practices.

Projects

- **CF Recommender Lab** (GitHub | Live Demo)
  - Developed a comprehensive Streamlit web application implementing **User-Based (UBCF) and Item-Based (IBCF) Collaborative Filtering algorithms**.
  - Features interactive visualizations, model evaluation metrics, and side-by-side performance comparison for recommendation systems.
- **AI Career Roadmap Recommender** (GitHub)
  - Built an **AI-powered decision support system** for Kenyan students, utilizing **NLP** for career matching and automatic eligibility checking.
  - Generates personalized learning roadmaps based on job market demand and KUCCPS requirements.
- **KUCCPS Dashboard** (GitHub)
  - Created an interactive Streamlit dashboard for analyzing student placement data, featuring **trend analysis, institutional comparisons, and program distribution**

insights.

- **Loan Default Prediction** (GitHub)
  - Developed a **machine learning model** for predicting loan default risk, implementing multiple classifiers with hyperparameter tuning and comprehensive evaluation metrics.
- **VendorHub – Multi-Vendor E-Commerce Platform**
  - Designed and developed a scalable **full-stack e-commerce system** using **React, Node.js, Express, and MongoDB**.
  - Implemented secure **RESTful APIs** for authentication and product management.

Education

Bachelor of Science in Data Science | Meru University of Science and Technology | Meru, Kenya*Expected Graduation: October 2026*

Nyambaria boys high school**KCSE B-**

Certifications

- **Python (Basic)** | HackerRank | 2024
- **SQL (Basic)** | HackerRank | 2024
- **Machine Learning** | Coursera | 2025
- **AWS Cloud Practitioner** | Amazon Web Services | 2025